Introducing 10 km/h in Dresden‘s inner city

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1 Introduction

MATSim is an agent-based transport simulation framework which can simulate the implementation of policies or infrastructure modifications on a traffic network. Agents representing a real population are assigned activity chains, they make travel plans to fulfill the chains and execute those on a given traffic network. After an iterative optimization process of performing, scoring and replanning the agents plans are output. These make up the simulated traffic [1].
The 1% Dresden scenario simulates traffic in the Dresden area. It consists of a synthetic agent population 1% the size of the real population of the area and a network with 1% of the real capacity.

2 Purpose & Research Question

Implementing extensive low speed traffic zones in urban contexts is gaining popularity all over Europe [3,4]. Traffic simulation software like MATSim can be used to gain an understanding of the effects of implementing such policies before realizing them. In this examination a low speed traffic zone was implemented in Dresden's downtown. 557 network links - shown in Fig. 1 - were assigned a freespeed value of 10 km/h using the changeset tools provided by Tramola.
While introducing such a large scale 10 km/h zone in the inner core of Dresden is probably not politically feasible the simulation can provide insight into effects of such a policy.
Three research hypotheses are listed below:

1. Rerouting: Tempo-10 raises travel costs, prompting zone avoidance - strongest among through traffic with no locational constraint.
2. Travel Effort: Travel time increases; distance may shift in either direction as MATSim optimises generalised cost, not route length.
3. User Groups: Through traffic reroutes more readily; destination traffic and residents remain due to fixed activity locations.

Fig. 1: Map of Dresden network links. Yellow links have been assigned freespeed of 10 km/h. Own illustration created with Tramola [2]

3 Methodology

Agents are considered directly affected if they completed at least one motorised leg (car, bike, ride, truck) with a route passing through one of the 557 changeset links in the Base Case. Identification is based on the complete link sequence stored in the <route type="links"> element of the output_experienced_plans.xml.

Base Case routes reflect pre-policy behaviour and indicate who would have used the zone under normal conditions. The comparison group in the Policy Case consists of the same person IDs, whose routing behaviour may change in response to the speed reduction

Affected agents can be classified into subgroups based on their activity locations:

Group A: Through Traffic: Origin and destination are both outside the zone
Group B: Destination Traffic: ≥1 non-home activity located on a changeset link
Group C: Residents: Home activity located on a changeset link.

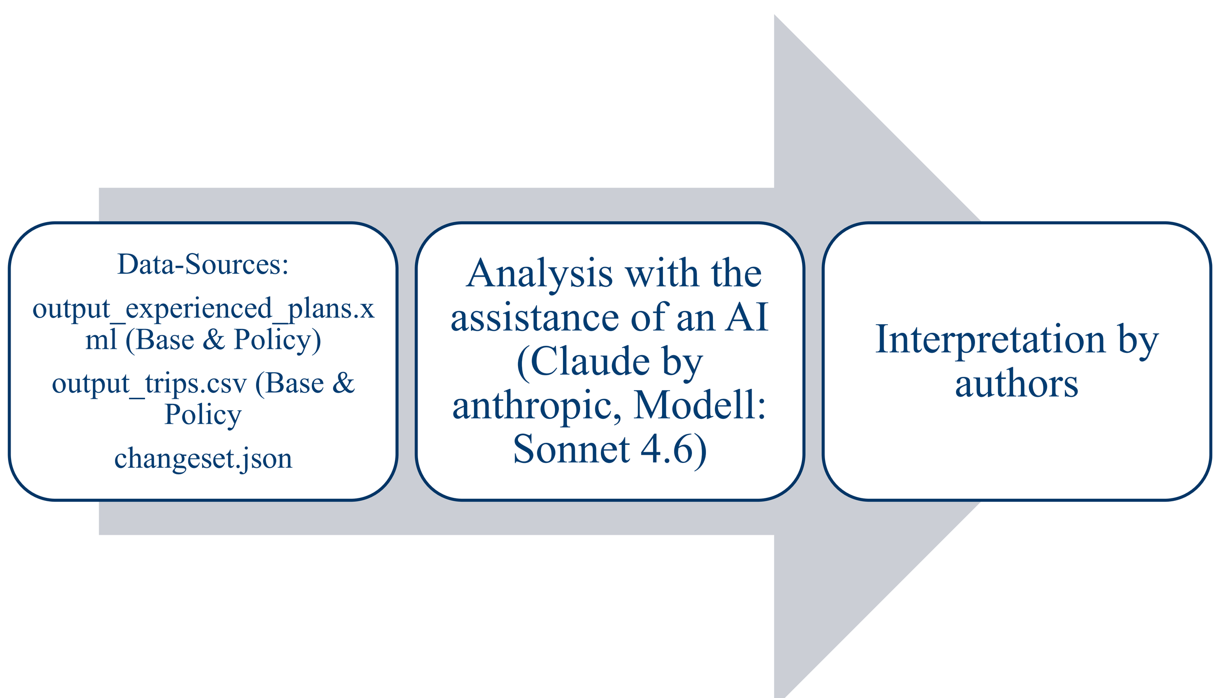

Fig. 2: Three-Step-Analysis. Own illustration

4 Results

The model simulates 8,942 persons in the 1% sample. Of these, 649 agents (7.3%) travel through the Tempo-10 zone at least once in the Base Case – extrapolated to approximately 64,900 persons in the real world.

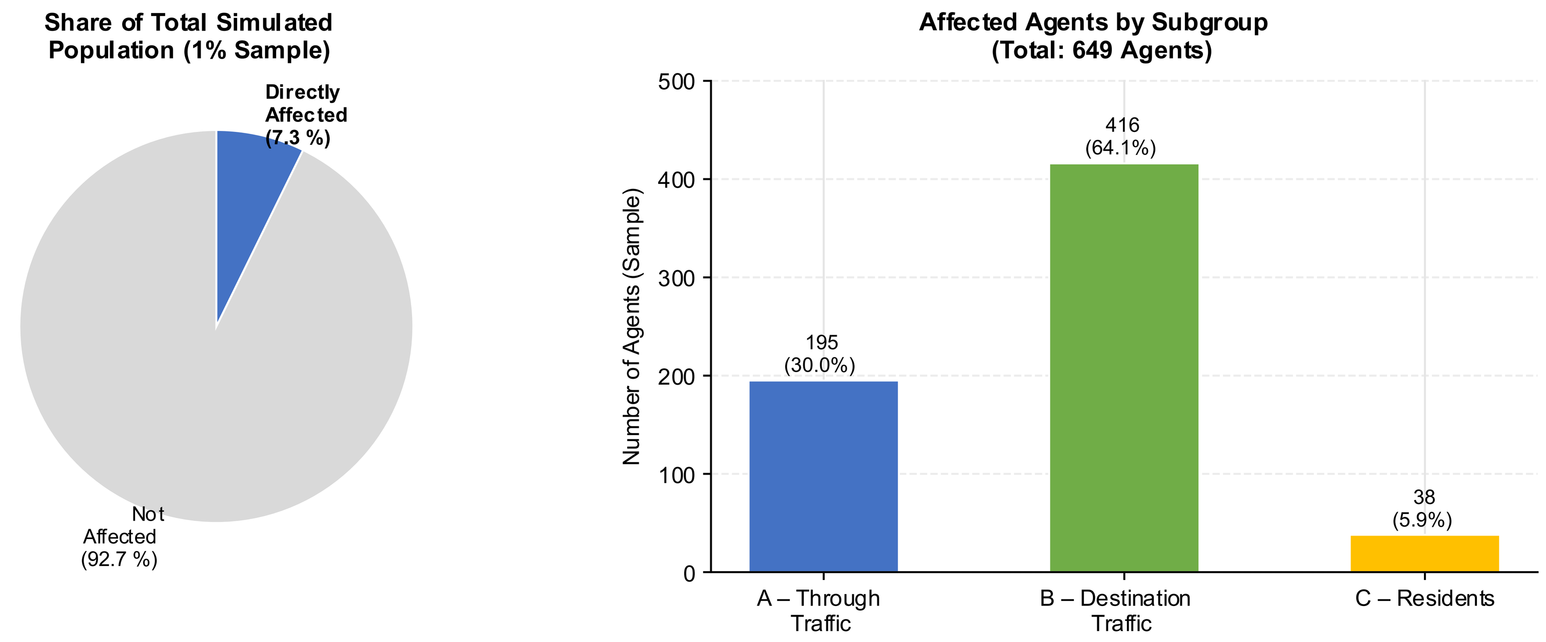

Fig. 3: Overview of directly affected agents. Own illustration created with the help of AI (Claude, Model: Sonnet 4.6)

Total distance travelled by affected agents decreases by 125 km in the sample (extrapolated: −12,500 km). Travel time increases by 47.85 hours in total (extrapolated: +4,785 h). The number of trips remains unchanged – no agent has fundamentally restructured their daily plan.

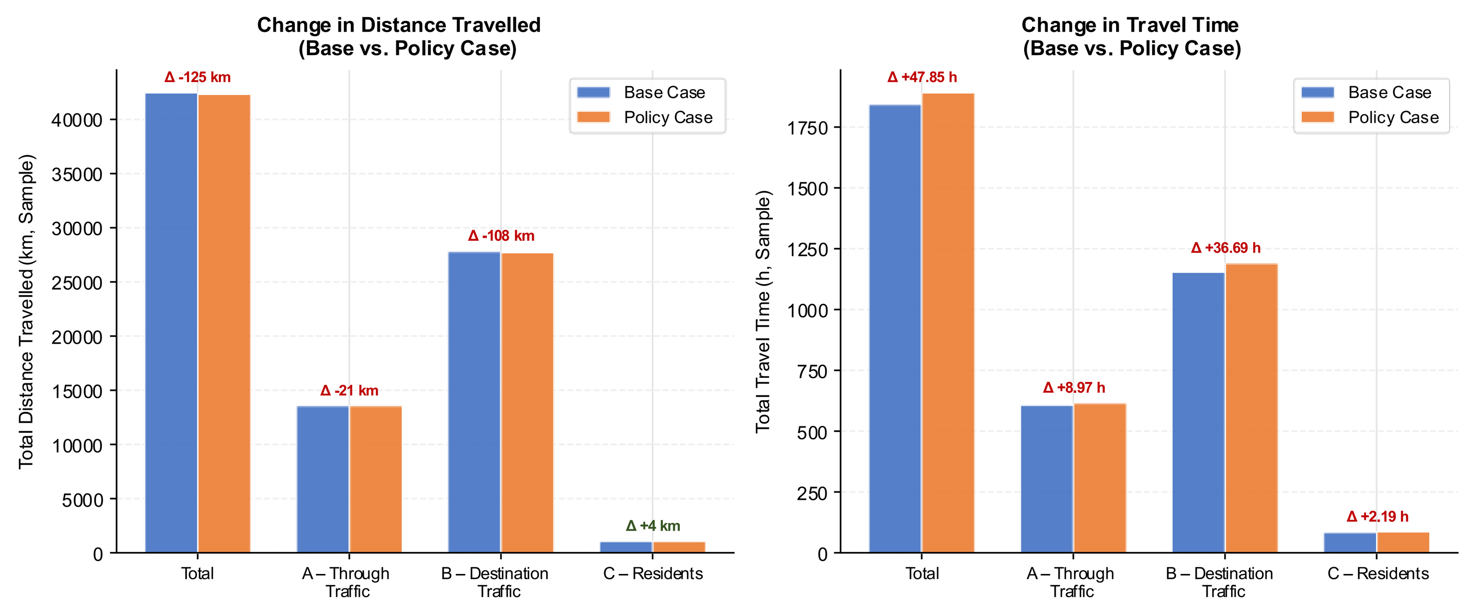

Fig. 4: Distanced Travelled & Travel-Time. Own illustration created with the help of AI (Claude, Model: Sonnet 4.6)

In the Policy Case, 605 agents still travel through the zone (compared to 649 in the Base Case). Of the 649 Base Case agents:

- 546 agents (84.1%) remain in the zone (Stayers) - predominantly destination traffic (95%) and residents (89.5%).
- 103 agents (15.9%) leave the zone (Leavers) - especially through traffic (40% of Group A).
- 59 new agents appear in the zone in the Policy Case (New Entrants)

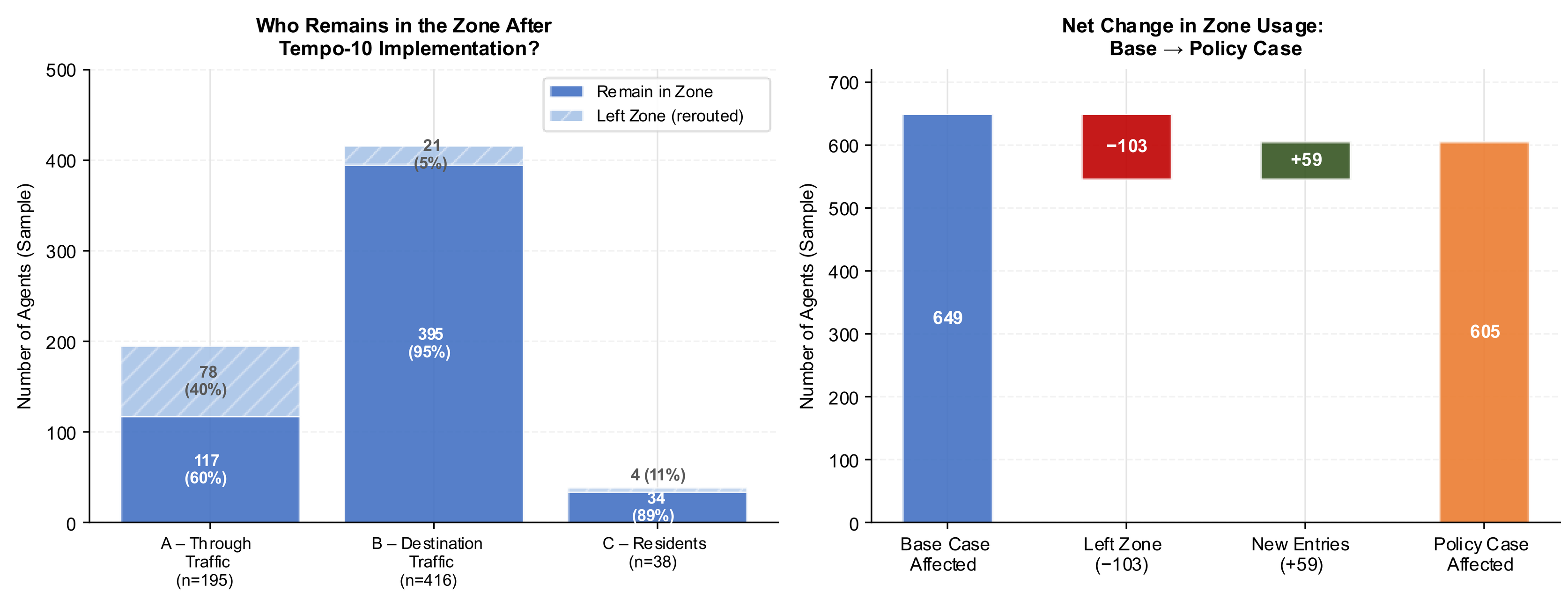

Fig. 5: Who remains in/ leaves the zone? Own illustration created with the help of AI (Claude, Model: Sonnet 4.6)

5 Conclusion

649 agents (7.3%) are directly affected

1. 40% of through traffic left the zone; yet 84% of all affected agents remained, limiting the overall deterrent effect.
2. Travel time rose (+47.9 h, sample); distance fell slightly (−125 km), suggesting some agents adopted shorter but slower alternative routes
3. Destination traffic (95% remain) and residents (90%) show strong locational inertia; through traffic shows the clearest avoidance response.